Finals Physics Review Practice Answer

MCQ

1. C
2. D
3. D
4. D
5. D
6. D
7. C
8. D
9. A
10. A
11. B
12. A
13. C
14. D
15. C
16. D

True and False

1.True
The right-hand rule relates current direction to the magnetic field in a solenoid.

2.True

3.False

4.False

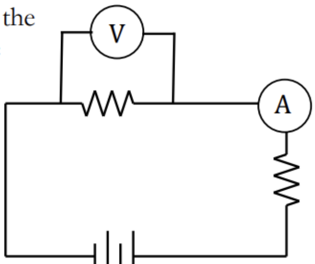
Calculation

1.

2. $1.4\times10{-}6$ T into the page B is 3% of BE

3. near: $4.4\times10{-}2$ N/m attractive

a. The voltmeter must be placed *over* the resistor to measure the potential difference before and after.



b. The ammeter must be inserted *into* the circuit so that there is only one path for current to follow. The ammeter may be placed anywhere in the circuit to measure current because it's a series circuit and the current is the same everywhere. But it make s most sense intuitively to place the meter near the resistor.

c. The two resistors are in series, so the current I is the same throughout the circuit. Using Ohm's Law and the information we have for R_1:

$$V_1 = IR_1$$
$$3V = (0.005A)R_1$$
$$R_1 = \frac{3}{0.005} = 600\Omega$$

d. Using the stated current and resistance:

$$P = I^2R$$
$$P = (0.005A)^2(1000\Omega) = 2.5e-2W = 25mW$$

e. The terminal voltage of the battery can be calculated using the total I and R for the circuit, along with Ohm's Law:

$$V = IR$$
$$V = (0.005A)(600\Omega + 1000\Omega) = 8.00V$$

f. Terminal voltage is emf less the potential drop due to the internal resistance:

$$V = \varepsilon - Ir$$
$$8 = 8.2 - (0.005)r$$
$$r = 40\Omega$$

far: 2.2×10−2 N/m repulsive